

Autonomous Pancreatic Lesion Detection and Radiologist Review System

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DEGREE	B.S. Applied Artificial Intelligence and Data Engineering
DATASET	PanTS (Johns Hopkins University) and PANORAMA (Radboud UMC / UMC Groningen)
STATUS	Draft for review

1. What? — Project Summary

Project Summary

The problem. Pancreatic cancer has a five-year survival rate below ten percent, and the largest single reason is late detection. The pancreas is difficult to work with on CT: it is soft tissue with poor contrast against everything around it, it varies widely in shape and position, and early tumors are small and easy to miss on a scan acquired for an unrelated reason. Published studies show that radiographic signs often appear months before diagnosis and go unnoticed. This is not a failure of attention — it is a hard perception problem inside a high-volume workflow.

The user. The direct user is a radiologist or radiology clinician reviewing abdominal CT. The indirect beneficiary is the patient, for whom earlier identification of a suspicious region changes the available treatment options. **This system does not diagnose and makes no clinical claim.** It proposes regions of interest that a qualified human accepts, edits, or rejects, and it records that decision.

What it is. A CT volume arrives and the system runs without further input: it locates the pancreas on the full scan, segments the organ and any lesion, scores the patient for tumor presence, and computes measurements — lesion volume, maximum diameter, anatomical subregion. Those findings populate a review interface where a clinician sees the scan with overlays, a drafted set of structured findings, and a locally hosted knowledge assistant that answers questions using retrieval over indexed pancreatic imaging literature. Every statement the assistant makes is tied to a source shown beside it; when retrieval finds nothing relevant, it declines rather than generating. The reviewer's accept, edit, or reject decision is stored, which means the system produces a corrections record as a byproduct of being used. Full architecture, data flow, and schema are in the technical appendix.

How this exceeds prior coursework. The five-week machine learning project preceding this produced a working segmentation model, a serving endpoint, and a review interface. It has one disqualifying limitation: the model is handed the pancreas location before it segments. Every number it produces — lesion Dice

0.474, detection sensitivity 96 percent, specificity 17 percent on the official 901-scan held-out test set — assumes something already found the organ. Published models on the same benchmark find it themselves, so the results are not comparable and cannot be submitted anywhere. The capstone removes that assumption. It also adds a second dataset requiring genuine cross-source integration, a relational and vector data layer, a separate retrieval-augmented AI system, and an external validation path.

Professional quality and feasibility. The existing repository carries a registered model with a version and checksum, twenty-seven pre-registered experiments with recorded decisions, two independent code audits, and complete documentation. The capstone adds a model card, data lineage, a validation suite that runs on ingest, and a review session with a practicing clinical stakeholder. Because the model, serving layer, and interface already work, each pillar is scoped with a committed minimum and an optional stronger version, so a difficult week costs sophistication rather than a component. System capabilities are committed and must work; experimental outcomes are pre-registered and reported honestly either way.

2. Why? — Project Motivation and Capstone Value

Project Motivation and Justification

I chose the pancreas because it is hard. It is soft tissue, it hides in a crowded abdomen, and pancreatic cancer is difficult to see even when someone is looking directly at it. My aunt is a clinician's assistant who reviews scans and then sits with patients to explain what the doctors found — including telling them they have cancer. When I described this project to her, she told me the pancreas is an organ radiologists do not enjoy imaging. That conversation shaped the problem statement more than any paper I read. Software that quickly finds the organ, outlines it, and flags a region worth a second look does not replace her judgment or anyone else's. It gets a human to the right part of the image faster.

There is a lot of healthcare behind me. My step-grandfather was a radiologist his entire career, and my mother started radiology school. My grandfather was a vice president at Intermountain Healthcare and oversaw the Urban South region. He read research constantly and tried to bring cutting-edge work into the hospitals he was responsible for, and a great deal of what he read came out of Johns Hopkins. That is close to exactly what I am doing now. The dataset I train on is Johns Hopkins' PanTS. The pretrained model I fine-tune is their SuPreM checkpoint. I am taking research from the institution he followed and trying to turn it into something operational. Johns Hopkins is also where I hope to attend graduate school, so choosing a dataset released by the lab whose work I want to build on was deliberate rather than incidental.

The technical reason for this dataset is just as concrete. PanTS has an active external evaluation path: the authors accept submitted models and score them on 26,489 held-out scans from UCSF, a Polish cohort, Peking University, and an RSNA dataset re-annotated by their team. Almost no public dataset offers that. It means the work can be measured against something I cannot see, cannot overfit, and did not grade myself.

The unmet need is specific. My current system finds tumors well — 96 percent patient-level detection, higher than either published model on the benchmark — but flags most healthy patients as suspicious. A tool with that false-positive rate is not usable, and fixing it is a real engineering problem rather than a cosmetic

one. I want to finish this because I want it to actually work, and because I believe the best use of AI is in healthcare, where a small improvement in how fast someone looks at the right image is worth more than almost anything else I could build.

3. How? — Technologies and Applied AI Skills

Technologies and Skills

- **Python, PyTorch, and MONAI — 3D medical image segmentation**

Demonstrated capability: Training a 3D SegResNet by transfer learning from a pretrained abdominal CT checkpoint; volumetric preprocessing (isotropic resampling, Hounsfield windowing, ROI extraction); sliding-window inference over full volumes.

Current experience level: Solid, used extensively in the preceding project.

Learning plan: No new tooling. The growth is in the problem — replacing a provided region of interest with a predicted one changes the failure modes entirely.

- **PostgreSQL with pgvector — unified relational and vector storage**

Demonstrated capability: A schema in which cohort membership is a constrained relation rather than a text file, making the leakage that previously invalidated my results impossible to express; hybrid queries combining semantic similarity with structured predicates.

Current experience level: Solid with PostgreSQL and SQL; new to pgvector specifically.

Learning plan: pgvector is an extension over familiar ground. I will build the index and tune HNSW parameters against a held-out retrieval question set in Week 6.

- **Retrieval-augmented generation with a locally hosted LLM (Ollama, Qwen)**

Demonstrated capability: A grounded assistant that answers only from retrieved passages, cites each visibly, and declines when retrieval finds nothing relevant. Running locally means no imaging-derived data leaves the machine.

Current experience level: Solid — I have built embedding pipelines, vector search, and RAG systems several times and already run local models.

Learning plan: The new work is evaluation rather than construction: recall@k and mean reciprocal rank for retrieval, and measuring what fraction of generated claims trace to a source.

- **Cross-source data integration and entity resolution**

Demonstrated capability: Merging PanTS with PANORAMA despite different label conventions, annotation provenance, reference standards, and tumor-type coverage, and identifying the cases that appear in both using institution matching and tumor-volume distribution fingerprinting.

Current experience level: New. I have never merged two clinical imaging datasets.

Learning plan: Build the exclusion list and label remapping behind unit tests and a load-time assertion, using the same guard pattern that now protects my training splits.

- **MLflow — experiment tracking and model registry**

Demonstrated capability: Versioned model registration with checksums; every run tied to its cohort, preprocessing configuration, and git revision.

Current experience level: Solid.

Learning plan: Extend to a PostgreSQL tracking backend so local and cloud runs share one store.

- **FastAPI, React, and NiiVue — serving and volumetric review interface**

Demonstrated capability: An inference endpoint triggered on scan arrival; a browser-based tri-planar and 3D viewer with overlays, measurements, and a persisted review decision.

Current experience level: Solid with FastAPI and React; NiiVue used once, in the preceding project.

Learning plan: Deepen NiiVue for measurement display and comparison views from its documentation.

- **Docker and rented cloud GPU compute**

Demonstrated capability: A pinned environment that runs identically on an Apple Silicon laptop and a rented NVIDIA instance, raising throughput from one overnight run to several per day.

Current experience level: Solid with Docker; limited with rented GPU infrastructure, as my cloud ML experience is with Azure.

Learning plan: Start on-demand rather than preemptible, re-establish a matched baseline on the new hardware before drawing any comparison, and hold numerical precision constant so results stay comparable to prior runs.

- **Experimental methodology — pre-registration and honest reporting**

Demonstrated capability: Every hypothesis recorded with a single changed variable and an accept-or-reject threshold written before the run; paired per-case comparison with bootstrap confidence intervals; results reported at the pre-registered horizon regardless of outcome.

Current experience level: Solid, developed during the preceding project.

Learning plan: Extend the same discipline to retrieval quality, the part of this project most at risk of being judged by impression rather than measurement.

4. Ten-Week Project Schedule

Week 1 begins Monday, October 6.

Week	Primary Goal	Planned Work and Milestones	Measurable Deliverable	Ahead/Behind Indicator
1	Relational foundation	Deploy PostgreSQL schema; ingest the PanTS manifest; convert cohorts from text files to constrained tables; write the ingest validation suite	9,901 scans indexed; cohort disjointness enforced by database constraint; validation suite passing	Ahead: PANORAMA download started. Behind: cohorts still defined by text files
2	Second dataset integrated	Download PANORAMA; unit-test the label remap; automate exclusion of the 274 overlapping Decathlon and NIH cases; build the merged cohort	1,964 usable cases ingested with provenance columns; entity-resolution report; merged cohort verified disjoint from evaluation sets	Ahead: EXP-27 smoke test run. Behind: remap or exclusion incomplete
3	Autonomy	Retrain the pancreas localizer on leak-free splits; measure effective containment after the final crop, not before it; run the cascade end to end	First autonomous lesion Dice on held-out PanTS, with no provided region of interest	Ahead: specificity gate prototyped. Behind: containment still unmeasured
4	Pre-registered training experiment	Run EXP-27: does adding 578 non-overlapping tumor cases improve lesion Dice? Evaluate against the bar fixed before the run	Accept or reject decision with paired bootstrap confidence interval; result recorded either way	Ahead: a second configuration trained. Behind: run not finished at the pre-registered horizon
5	Usable operating point	Build the volume gate and tumor-presence classifier; select the threshold on validation and report on test	Specificity materially above 17% with detection sensitivity at or above 96%, or a documented account of why that trade is unavailable	Ahead: submission package drafted. Behind: no operating point beats baseline on both axes
6	Knowledge corpus	Build and pin a filtered PubMed and PMC Open Access snapshot; chunk and embed; build the pgvector HNSW index; author a retrieval evaluation set	Queryable index with recall@k and mean reciprocal rank reported on held-out questions	Ahead: embedding ablation started. Behind: index not queryable
7	Grounded assistant	Wire Ollama and Qwen to the retriever; implement cite-or-refuse; construct queries from the pipeline's own findings	Assistant answering with visible citations; measured groundedness rate; refusal verified when retrieval fails	Ahead: biomedical versus general embedding ablation complete. Behind: any ungrounded generation reaching the interface
8	Integration and automation	Trigger inference on scan arrival; build the suspicion-ranked worklist; surface overlays, measurements, and the assistant in the interface; persist review decisions	End-to-end demonstration from scan arrival to a recorded reviewer decision	Ahead: stakeholder session scheduled. Behind: components not wired end to end
9	Evaluation and stakeholder feedback	Final held-out evaluation; leave-one-institution-out analysis across sites; structured review session with a clinical stakeholder; incorporate feedback	Final metrics table with confidence intervals; per-site results; documented stakeholder feedback and the changes it produced	Ahead: Johns Hopkins submission package complete. Behind: final evaluation incomplete
10	Package and present	Model card, data lineage documentation, README; assemble the external submission package; build and rehearse the presentation	Complete documentation set, presentation delivered, submission package ready to send	Ahead: model submitted to Johns Hopkins. Behind: documentation incomplete

Stretch items, deliberately outside the critical path: external evaluation by the Johns Hopkins team; the biomedical embedding ablation; a style fine-tune for report formatting.

5. BSAAI Capstone Pillars

Available Pillars

- Data Analytics and Decision-Making
- **Data Engineering** — selected
- **Machine Learning** — selected
- Process Automation
- Data Science
- Strategic Technology Leadership
- Cloud Technology
- **Applied Artificial Intelligence** (*listed under Core Disciplines in the reference guide rather than in the checkbox list above — see note*) — selected

Note for the reviewer: I selected Machine Learning, Data Engineering, and Applied Artificial Intelligence based on guidance from my degree chair. Applied Artificial Intelligence appears in this document as a core discipline but not in the pillar checkbox list. If the three selections must come from the checkbox list, the closest mapping is Machine Learning, Data Engineering, and Data Science, with Process Automation and Strategic Technology Leadership both available as a fourth. I am happy to relabel.

Pillar 1: Machine Learning

How the project demonstrates this pillar:

The core of the project is a 3D segmentation and detection model trained by transfer learning from a pretrained abdominal CT checkpoint. The capstone covers the full lifecycle: replacing a provided region of interest with a learned localization stage, retraining on an expanded cohort under a pre-registered hypothesis, building a downstream classifier to fix a specific failure mode, selecting an operating point on validation and reporting it on held-out test data, versioned registration, and containerized deployment behind an inference endpoint.

The objective is measurable. The current model reaches 96 percent patient-level detection sensitivity but only 17 percent specificity, and its lesion Dice of 0.474 depends on being given the organ location. The capstone removes that dependency and attacks the specificity gap, with every experiment declared before it runs and reported whether it succeeds or fails.

Pillar 2: Data Engineering

How the project demonstrates this pillar:

This project merges two clinical imaging datasets that are not compatible: different label conventions, different annotation provenance, different negative reference standards, different tumor-type coverage, and different contrast phases. Producing a single trainable cohort requires a canonical schema, a tested per-source mapping layer, and provenance carried as data rather than assumed. It also requires entity resolution across sources — I have already shown that the Medical Segmentation Decathlon pancreas dataset sits inside PanTS, and that PANORAMA absorbs 194 Decathlon and 80 NIH cases. Merging naively would place training data inside the evaluation set. Details are in appendix section A4.

Alongside that sits feature engineering — the volumetric preprocessing pipeline, derived measurements stored with lineage, and the feature set consumed by the tumor-presence classifier — and data quality enforcement informed by defects already found in this data, including a validation split leak that inflated a headline result by 0.11 Dice before I caught it.

The data engineering is not parallel to the machine learning here; it is upstream of it. The Week 4 experiment cannot run at all until the integration and deduplication work is correct.

Pillar 3: Applied Artificial Intelligence

How the project demonstrates this pillar:

The second system is a retrieval-augmented assistant built on a locally hosted large language model, grounded in an indexed corpus of peer-reviewed pancreatic imaging literature. It is not a chatbot bolted onto a viewer: the imaging pipeline's structured output — lesion volume, maximum diameter, anatomical subregion, contrast phase — becomes the retrieval context, so the two systems are joined through the data layer rather than running side by side.

The design constraint is a safety property, not a disclaimer. The assistant may only make statements traceable to a retrieved passage displayed next to it, and must refuse when retrieval returns nothing relevant. That is testable, and it is what keeps a research tool from drifting into unlicensed clinical advice. Retrieval is measured with recall@k and mean reciprocal rank against a held-out question set, and generation by groundedness rate, so the AI component is held to the same evidentiary standard as the imaging model. The full pipeline specification is in appendix section A6.

6. Reflection

Personal Growth and Technical Resilience

The hardest part of the project that preceded this was not building the model. It was making it more accurate. You can run a hundred training runs and they do not simply get better. That realization changed how I work: I started writing down a hypothesis and a single changed variable before every run, along with the threshold that would count as success, and only then training. It is the biggest improvement I made, and if I could start over I would have done it from day one instead of week three. I am carrying it into the capstone from the beginning.

That discipline gets tested here in a way it has not been before. This project already has two rejected experiments, and one accepted result that turned out to be a false positive because I read the metrics before the run had finished. I have withdrawn a headline number of my own after finding a validation leak in my splits that inflated it by 0.11 Dice. Reporting that honestly cost me the best result I had. The capstone will produce more moments like that, and the plan explicitly separates committed system capabilities — which must work — from pre-registered experiments, which are allowed to fail and still count as results.

Technically I am moving into territory I have not been in. I have never merged two clinical imaging datasets with different annotation provenance and overlapping membership, and that is exactly where data leakage hides. I have never removed a human-provided region of interest from a pipeline built assuming one. I had not worked with 3D images at all before June, and I am taking a computer vision course concurrently, which is priming me directly for the localization work.

The largest source of uncertainty is one I do not control. The stretch goal is having my model evaluated by the research group that published the dataset. They may be slow, or the result may not be good. The project is deliberately built so that outcome changes nothing structural — I will have built an autonomous, externally validatable system either way, and I will report what comes back exactly as it comes back.